

Wyatt Walsh

SENIOR AI/ML ENGINEER

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Senior AI/ML engineer shipping production agentic AI, LLM document intelligence, retrieval workflows, AI guardrails, data platforms, and fintech tooling.

EXPERIENCE

Senior AI/ML Engineer

Frame Payments | Remote | Oct 2025 - Jan 2026

Delivered production agentic AI document intelligence, human-in-the-loop review, and multimodal compliance screening for high-risk merchant onboarding.

- Deployed production LLM document intelligence with PydanticAI, Amazon Bedrock AgentCore, and Claude; added human-in-the-loop browser feedback for confidential uploads and reduced analyst follow-ups/resubmissions ~90% by blocking miscategorized documents before review.
- Built multimodal restricted-item compliance screening by combining product images and structured metadata; translated banking-partner policy requirements into automated risk checks for merchant onboarding across tens of merchants and multi-location catalogs.
- Built a NotebookLM-based technical knowledge hub spanning vendor/web documentation and internal monorepo docs; automated corpus refresh with a Colab pipeline that re-ingests the latest sources to keep retrieval accurate and up to date.
- Drove internal AI adoption with Notion MCP playbooks, high-signal prompt patterns, and weekly office hours reaching ~15% of the company; standardized PII/confidential-document safeguards and auditability requirements.

Associate III, Senior Software Engineer - Data Integrations

JPMorgan Chase & Co. | New York, NY | May 2023 - Apr 2025

Built enterprise data pipelines, risk analytics, and data-quality automation for credit trading while scaling secure internal AI developer tooling.

- Pioneered firm-wide rollout of secure generative AI developer tools (LLMs, GitHub Copilot, Codeium) as one of the first 500 engineers, shaping scalable deployment and leading workshops for technical and non-technical teams.
- Devised ML-driven anomaly detection for credit trading and risk analytics, reducing false positives by 25% and manual review time for risk teams.
- Developed a resilient Python wrapper for KDB+ with CI-style tests and optimized Q functions, democratizing data access for 5,000+ users.
- Engineered a cloud-function-driven asset-fee data sink, consolidating 10+ sources into a centralized database, cutting manual maintenance by 90% and providing on-demand access for traders through a Python SDK, REST API, and interactive dashboard.

Senior Software Engineer

Stealth Web3 Startup | Remote | Mar 2022 - Jul 2022

- Architected a scalable collateralized lending app with React, Next.js, TypeScript, MobX, and ethers.js for secure wallet and smart-contract workflows.
- Implemented CI/CD with GitHub Actions, Husky, Cypress, and Jest, halving release cycles, raising test coverage to 90%, and mentoring a front-end intern.
- Optimized front-end performance through code splitting, lazy loading, and asset caching, reducing initial load times by 40% and JavaScript bundle size by 30%.

Founder, Chief Technology Officer

SandLabs | Remote | Jun 2021 - Mar 2022

- Launched a Next.js/TailwindCSS public site with top-tier Lighthouse scores and secured \$75K in competitive ecosystem grant funding.
- Built Python pipelines that normalized a 1M+ Reddit posts/comments dataset into a reusable analytics data product for external users.

Demand Forecasting Extern

Gap Inc. | San Francisco, CA | Jan 2020 - May 2020

- Applied time-series decomposition and bias correction techniques to demand forecasts, boosting accuracy by up to 25% and minimizing surplus inventory costs.
- Owned the data engineering layer by architecting scalable MySQL and Apache Spark pipelines across 5+ years of sales data, improving data processing throughput by 95% and standardizing insights across forecasting teams.

SKILLS

AI, LLM & AGENT ENGINEERING: OpenAI API, Anthropic Claude API, Amazon Bedrock, LangChain, LangGraph, DeepAgents, PydanticAI, Model Context Protocol (MCP), AI Agent Skills, Guardrails AI, Retrieval Augmented Generation (RAG), Prompt Engineering, Context Engineering, LLM Evaluation

DATA ENGINEERING, DATABASES & ANALYTICS: PostgreSQL, KDB+, qPython, DuckDB, BigQuery, AWS Athena, Apache Spark, AWS Glue, Apache Airflow, Prefect, Polars, SQLAlchemy, Pydantic, Pandera, Parquet, ETL, Data Validation

STREAMING, MESSAGING & CAPITAL MARKETS SYSTEMS: Publish/Subscribe Architecture, AMPS, Apache Kafka, Amazon Kinesis, Google Cloud Pub/Sub, Event-Driven Architecture, Market Data Systems, Low-Latency Messaging

CLOUD, DEVOPS & OBSERVABILITY: AWS, GCP, Docker, Kubernetes, Terraform, GitHub Actions, AWS Lambda, Vercel, Prometheus, Monitoring, CI/CD

LANGUAGES & AUTHORING: Python, SQL, TypeScript, Go, Rust, q, Bash

PROJECTS

AI Agent Harness Configs | Live site | github.com/wyattowalsh/agents

Jan 2026 - Present | Python, TypeScript, MCP, FastMCP
Open-source agent tooling catalog and docs portal packaging 112 portable skills, 8 reusable agents, 30 MCP tools, and cross-runtime workflows for Claude Code, Codex, Copilot, Cursor, and related coding assistants.

- Standardized reusable agent definitions across Claude Code, Codex, GitHub Copilot, Cursor, OpenCode, and related IDE agents.
- Shipped generated docs, install flows, and 30 MCP-ready tool presets spanning code review, security, infrastructure, research, and workflow automation.
- Packaged 112 portable skills and 8 reusable agents into a cross-runtime workflow system for AI-assisted software development.

NBA Basketball Database | Docs | Kaggle | github.com/wyattowalsh/nbadb

Apr 2021 - Present | Python, Pandas, Polars, SQLAlchemy, DuckDB, Pandera
Public NBA Kaggle dataset and repo covering 65K+ games and 13M+ plays; reached 425K+ views and 60K+ downloads before freezing at the 2023-07-06 update for reproducible sports analytics workflows.

- Implemented daily Pandera-validated NBA ETL plus DuckDB, SQLite, Parquet, and CSV exports.
- Building the next-generation nbadb rebuild with 152 extractor wrappers and 204 generated public star-schema outputs.
- Published a public sports analytics dataset with 65K+ games and 13M+ play-by-play events across multiple warehouse formats.

ProxyWhirl | Live site | github.com/wyattowalsh/proxywhirl

May 2025 - Present | Python, SQLite, PyPI
Released rotating proxy toolkit with 9 strategies, 100+ auto-fetched sources, self-healing circuit breakers, and five interfaces for scraping/data workflows.

- Implemented 9 proxy rotation strategies with configurable fallback chains, per-source scoring, and performance-based selection.
- Built self-healing validation across 100+ proxy sources with circuit breakers, auto-ejection, recovery promotion, and workflow-specific health scoring.
- Released the toolkit for resilient scraping and data-collection workflows with SQLite-backed state and PyPI distribution.

AI/ML Web Feeds | Live site | github.com/wyattowalsh/ai-web-feeds

Oct 2025 - Present | Python, Pydantic, SQLModel, YAML, Fumadocs, Vercel
Built a curated AI and ML feed platform with typed schemas, HTTP/RSS validation, contributor submission workflows, OPML exports, and documented browse/download surfaces for research discovery.

- Defined YAML feed schemas with topic and source-type validation across blogs, papers, and newsletters for reliable feed quality.
- Shipped helper scripts, CI checks, scheduled validation jobs, and a docs-backed submission/review flow for reliable community feed curation.
- Published OPML exports and documented browse/download surfaces for AI and ML research feeds, downstream readers, and offline subscription managers.

MCP-Crawl4AI | Live site | github.com/wyattowalsh/mcp-crawl4ai

Apr 2025 - Present | Python, FastMCP, Crawl4AI, Playwright, Docker
Released FastMCP server wrapping Crawl4AI so AI agents can scrape, crawl, manage sessions, and retrieve artifacts through a focused MCP surface with documented stdio and HTTP integration paths.

- Built a FastMCP v3 server with managed headless Chromium, session-aware crawling, and automatic Playwright setup using two runtime dependencies.
- Exposed scrape, crawl, session, and artifact-retrieval workflows with docs, tests, and release-tagged MCP install flows.
- Documented stdio and HTTP integration paths so AI agents can use web crawling through a focused MCP surface.

Personal Website: w4w.dev | Live site | github.com/wyattowalsh/personal-website

Nov 2018 - Present | Next.js, React, TypeScript, Tailwind CSS, MDX
Long-running personal web app and MDX publishing system built with Next.js, pairing a polished public-facing frontend, a richly tooled technical blog, strong SEO, and durable content workflows.

- Maintained the public site across multiple generations, covering responsive landing pages, themes, motion, accessibility, and MDX publishing.
- Built custom MDX publishing with search, metadata, math rendering, feeds, tags, and integrated authoring tools.
- Sustained a polished public-facing frontend with strong SEO and durable content workflows.

EDUCATION & CERTIFICATIONS

Bachelor of Science
University of California, Berkeley - College of Engineering, Industrial Engineering and Operations Research (IEOR)
Aug 2014 - Dec 2020

High School Diploma
The Hotchkiss School
Aug 2012 - May 2014

AWS Certified Cloud Practitioner
Amazon Web Services (AWS) | Nov 2024

IBM Data Science Professional Certificate
IBM | Feb 2021